PRM: The Phonic Resonance Matrix — Concepts as Resonant Cavities and Inference as Wave Propagation

Irfan Mahir — Independent Researcher, Dhaka, Bangladesh — irfan@furylogic.com

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"Inhibition is not an `if` *statement. It is two waves arriving out of phase."*

Abstract

We specify a substrate in which a concept is not an identifier, a token or an embedding, but a **physical resonant cavity** with mass, displacement and velocity. Relations are not pointers but **tunnels** that transmit mechanical force between cavities. Inference is not search but **wave propagation**, and the answer to a query is whichever cavity is vibrating hardest once the field has settled.

The architecture's distinguishing property is that several capabilities usually requiring explicit algorithms become consequences of the medium. Inhibition is destructive interference rather than a conditional. Multi-constraint search is two wavefronts colliding rather than a set intersection. Structural analogy is sympathetic resonance between cavities that share a harmonic ratio and no edge at all.

Section 7 reports what was measured and what remains an argument from mechanism, including one claim in the original formulation that does not survive scrutiny.

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1. The Substrate

Concepts are **resonant volumes**. Each cavity carries a mass, a displacement x, a velocity v, and a natural frequency determined by its physical parameters. In the acoustic reading, a large cavity has a low resonant frequency and a small one a high frequency.

Relations are **physical tunnels** — channels connecting cavities, not references. Two polarities exist:

- An **excitatory** tunnel transmits displacement directly, producing sympathetic resonance in the target cavity.
- An **inhibitory** tunnel inverts phase by 180°, so its contribution actively cancels vibration at the target.

The design commitment is that the state of a concept is its *actual mechanical state*, not a value standing for one.

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2. Dynamics

The tick integrates Hooke's law and Newton's second law across every tunnel:

$$F \;=\; k\,(x_a \cdot p - x_b) \;-\; c\,(v_a - v_b)$$

$$v \;\mathrel{+}=\; \frac{F}{m}\,\Delta t, \qquad x \;\mathrel{+}=\; v\,\Delta t$$

where k is tunnel stiffness, c damping, $p \in \{+1, -1\}$ polarity, and forces are applied equal and opposite to the two endpoints.

A query is an injected driving force rather than a string. Reading the result means observing which cavities have the largest displacement after the field settles — the physical analogue of placing pressure sensors in the pipes and seeing which are shaking.

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3. Why Inhibition Requires No Logic

In a conventional implementation, negation is a branch: *if the relation is inhibitory, subtract instead of add.* Here the polarity term p sits inside the force expression, so an inhibitory tunnel produces a 180°-shifted contribution by arithmetic alone.

The consequence is architectural rather than aesthetic. Because inhibition is multiplication by -1 rather than a conditional, the update path contains no branch on relation semantics — which is what permits the whole tick to be expressed as scalar arithmetic and saturating nonlinearities, with no control flow that depends on what a relation *means*.

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4. Emergent Capabilities

These are the paper's substantive claims: behaviours obtained from propagation rather than from an algorithm written to produce them.

4.1 Constructive interference as multi-constraint search

Drive two cavities simultaneously — say `type_2_diabetes` and `autophagy`.

A graph search attempts to trace a literal edge path between them and fails if none exists. The matrix instead radiates waves outward from both. The wavefronts split, reflect, and meet. At a distant cavity — `mtor` — the two arrive **in phase**, amplitudes add, and that cavity spikes.

The system did not know `mtor` was the bridge and did not search for it. It vibrated because it sits at the geometric intersection where two independent wavefronts collided. The conjunction was computed by the medium.

4.2 Sympathetic resonance as structural analogy

Strike a 440 Hz tuning fork and a nearby string tuned to 880 Hz begins to sound, with no mechanical connection between them.

If cavity parameters reflect structural properties, then energy injected into one structure can excite another that shares a harmonic ratio **and has no connecting edge**. That is a structural analogy — a repurposing candidate — found through harmonics rather than traversal.

4.3 Standing waves as attractor discovery

Run the tick for a few thousand iterations and kinetic energy settles into a standing wave: some cavities become nodes of near-zero motion through destructive interference, others antinodes of maximum vibration.

Standing wave patterns are **computationally irreducible**: one cannot predict where the pattern forms by inspecting the rules; the simulation must be run. When the field settles, it has resolved the equilibrium of a large number of competing constraints simultaneously, and the high-amplitude cavities are the result.

4.4 Detecting this without a classifier

Novelty is observed rather than classified. After the field settles, scan for cavities with high amplitude that were **not** directly driven:

```
// After the wave has propagated, find the novelty
for (const auto& cavity : cavities) {
    if (cavity.displacement > THRESHOLD) {
        if (!was_injected(cavity.id)) {
            // A cavity is vibrating that we never excited directly:
            // an emergent resonance point.
            report(cavity.physical_label, cavity.displacement);
        }
    }
}
```

There is no `if/else` classifier deciding what counts as interesting. The condition is *energy arrived here and we did not put it here*, which is a statement about the field rather than about a category.

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5. Position Against Pattern Matching

Property	Neural / graph search	PRM
Concept state	activation value or embedding	physical displacement and velocity
Inhibition	a rule or conditional	mechanical cancellation of opposed waves
Reachability	programmed paths or learned weights	radiated energy; waves collide unprogrammed
Multi-constraint	explicit set intersection	interference of two wavefronts
Analogy	similarity metric over representations	sympathetic resonance at harmonic ratio
Output	argmax over scores	whichever cavity is physically shaking

The claim is that energy in a physical medium **radiates** rather than following single logical lines, and that novelty arises from the unprogrammed collisions of those radiating waves.

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6. Implementation Note

The substrate is a one-dimensional wave simulator of the kind used in game audio and fluid engines — tension, mass and displacement integrated directly. No tensor library, no autodiff, and no linear-algebra framework is involved, because nothing in §2 is a matrix operation.

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7. What Was Measured, and What Was Overstated

Implemented and load-bearing. Cavities with mass, displacement and velocity; polarity-carrying tunnels; the force integration of §2; and amplitude readout are the Layer-1 substrate of the current engine (`struct Cavity`, `physics.hpp`). The branchless property of §3 holds: the tick contains one guard branch and no conditional on relation type.

Measured. Attractor settling (§4.3) and the undriven-cavity novelty probe (§4.4) run and produce results. Multi-source interference (§4.1) occurs and drives the analogical bridging mechanism reported separately with 20-seed reproducibility data.

Overstated in the original formulation. Section 4.2's sympathetic resonance was presented as though cavity volume already encoded real structural properties. It does not. Cavity parameters are derived from graph position and relation fan-out, **not** from molecular structure, so the harmonic ratio between two cavities carries no independent structural information about the entities they denote. The mechanism is sound; the specific claim that it discovers *chemical* analogies without edges is unsupported by anything implemented, and analogical bridging in the current engine works through shared-target intersection rather than harmonics.

Not implemented. The literal acoustic framing — transducers, piezoelectric sensors, physical pipes — is expository. The substrate is a numerical simulation, and no claim is made that a physical acoustic device was built.

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8. Limitations

Settling is assumed. Section 4.3 assumes the field reaches a standing wave. A driven system may instead lock into a ground state and refuse to leave, which was later observed as a real failure mode requiring a dedicated escape mechanism.

Amplitude is not relevance. High displacement means energy arrived, which is not the same as being an answer. A highly-connected hub accumulates energy from every direction; this required a separate thermodynamic mechanism to suppress ubiquitous participants.

Cost scales with the field, not the query. Every tick integrates every tunnel, so a query costs the same whether it touches three concepts or three thousand. At large ontology sizes this dominates, and sparse iteration over only the active wavefront is the necessary and currently incomplete remedy.

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9. Conclusion

PRM's contribution is the identification of a representation in which several operations normally requiring algorithms are instead consequences of propagation: inhibition as phase inversion, conjunction as interference, and constraint resolution as settling.

The parts that survived are those making no assumption about what the cavities *denote* — the dynamics, the polarity treatment, the readout. The part that did not survive assumed cavity parameters encoded external structure they were never given, which is a useful reminder that a physical metaphor stops being explanatory exactly where the physics stops being connected to the domain.

References

- Mahir, I. *MIMS: Mechanical Impedance and Material Stiffness.* — the material table parameterising these tunnels.
- Mahir, I. *HNCA: Hierarchical Neural Constraint Architecture.* — the macro-layer wrapping these cavities into frequency-separated regions.
- Mahir, I. *RB+G: Gated Recursive Bifurcation with Analogical Bridging.* — the mechanism that actually performs analogical discovery.
- Mahir, I. *From Artificial to Synthetic Intelligence.* §5.4c.
- Implementation: `src/physics.hpp` (`struct Cavity`, `struct Tunnel`, `tick()`).